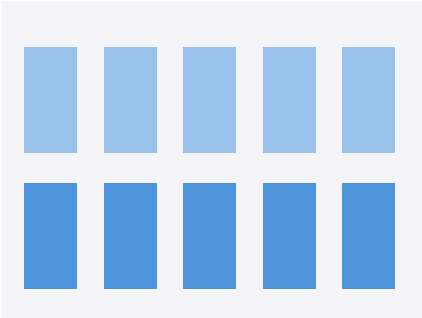
Shelfie

Paid only when the photo is true.

Isabel • Fernando • Lorenz • Helmut

TUM Blockchain Hackathon 2026 • Cardano track



[real shelf photo]

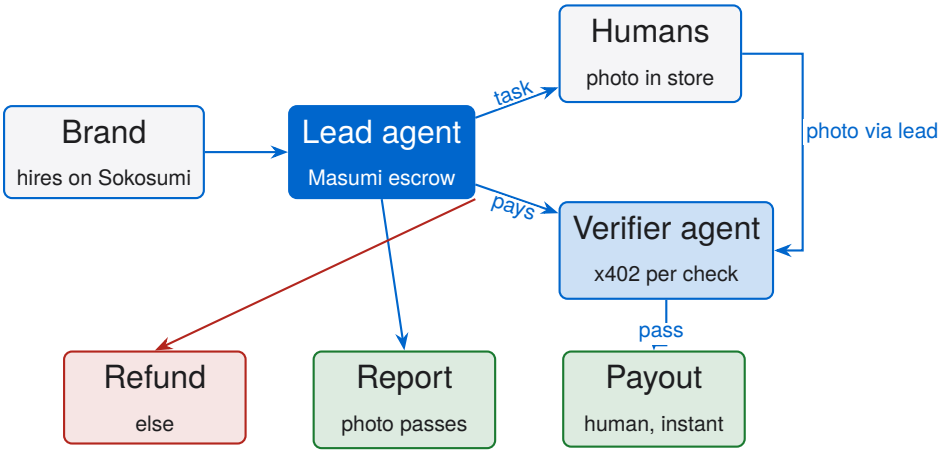
A friend of Isabel's father wants to bring a drink to Germany.

"Can you photograph the competitors in a few shops?"

What do they cost? Are they on the shelf?

A real request.

How it works


Humans
photo in store
task
Brand
hires on Sokosumi
Lead agent
Masumi escrow
photo via lead
pays
Verifier agent
x402 per check
pass
Refund
else
Report
photo passes
Payout
human, instant

LIVE DEMO

Shoot. Verify. Get paid.

The market

$1.1T
retail sales lost to empty shelves per year

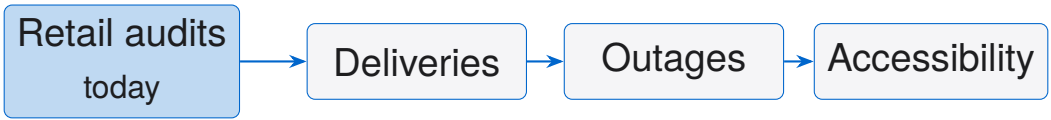
$6B
spent on in-store merchandising per year

$2.35B
mystery shopping, paid per visit

One check: €12, vs €50+ per test purchase.

IHL Group (2026 est.), Beroe (2023 est.), Straits Research (2025). Loss is not revenue; markets overlap and are not added. Shelfie prices are illustrative estimates.

Beyond the hackathon



AI agents can browse the web, but not see the world.

One engine: any proof-of-real-world task is a domain pack.

Roadmap; only retail checks are built. Next: store QR codes for location.

Agents hire people to look, and the blockchain pays the bill.

Shelfie

What is real today

Live on preprod

Escrow lock (Masumi)
Verifier fee over x402
Payout to the worker
One-time code, verifier gates

From an earlier run

Escrow settlement
Refund when no photo passes

The escrow cycle takes about an hour.

Not built yet

Location proof (store QR next)
Strangers as workers
Identity checks
Mainnet, real money

Demo: our own printed price tags, teammates as workers, a seeded Munich catalog.

The customer



A brand entering a market

No field team in the new country.

Needs before launch: competitor prices and availability, store by store.

Has: friends with phones, or an agency paid per visit.

One job = one store, one product, one verified photo.

The problem: retail execution

[X days]
booked ahead

€50+
per store visit

[XX%]
unusable data

Paid per visit whether or not the data is usable.

German Testkauf €50–260 (mystery.agency, mystery-shopping.de); US audit visit $30–45 (T-ROC Global 2026).

[TODO: source lead time and unusable share]

The idea

An agent hires people to photograph shelves.

It pays them only if the photo is verified true.

Three payment gates

1

Photo gate

Challenge code, price and product parse, in time window.

→ pays the human

2

Job gate

The passing photo is the job result.

→ escrow settles

3

Fallback

Otherwise the result is withheld.

→ automatic refund

Proof on Cardano preprod

Escrow lock



`[tx: lock]`

Human payout



`[tx: payout]`

Settlement / refund



`[tx: settle]`

Scan to open in the explorer.

Business model

Brand pays €12 per verified store

Human earns €3.50 per passed task

Verifier costs < €0.10 per check

Gross margin ∼65%

Report in `[XX min]` instead of days

Failed checks cost the brand nothing

€12 pay-as-you-go • €399/month for 50 checks • €5–6 at volume

Illustrative product pricing in EUR (estimates); not the preprod test amounts.

Why Cardano

Escrow Masumi holds the fee until delivery

x402 the verifier is paid per check, no accounts

Cheap fees micro-payouts to humans make sense

Receipts every check, payout and refund is a public tx

Team

Isabel	**Fernando**	**Lorenz**	**Helmut**
Agent and engine	Payments	Verifier	Web app and evidence